

ber of people, how wide it should be, where the wings need to be and how strong the materials should be.

As an electronics engineer, you can design the sensors and connections that tell the pilot of an aircraft that things are fine or that there is a problem. Mathematicians are also hired by the aerospace industry and if you are wondering why, these are the ones who develop the math formulae that engineers use to design their work. The physicist is another behind-the-scenes part of a design team, analysing a scientific problem for the aircraft such as overcoming the heat barrier or computing a trajectory.

Who's hiring?

Delving a little deeper, we can group major recruiters in this field under three broad categories. The first category is

Explains Prof. Sriram, "Aeronautical engineers deal with the design and analysis of aircrafts, while aerospace engineers deal with aircrafts and spacecrafts. In the Indian context, maintenance is also included, though traditionally, aviation (airline and airport operations) and maintenance are considered separate; to give an example, Tata and Volvo design and manufacture buses while the buses are operated and maintained by other road transport companies. The job categories are parallel to this—HAL, DRDO and ISRO design and produce aircrafts and spacecrafts, so these employ aerospace engineers to design and build these aircrafts and spacecrafts. Air India or other airlines operate the aircrafts and they have job opportunities in aircraft maintenance, which means keeping the aircraft in proper condition so it can be

many of these jobs are related to R&D, where higher qualification and experience are preferred. As a result, many of the graduates take up jobs that are neither in the core sectors nor related to the core field. However, some private Indian companies are coming up and various foreign and multinational companies are trying to expand their work in the field of aerospace in India. So the number of jobs will definitely increase in the near future.

Dr M.R. Patkar, director, Indian Institute for Aeronautical Engineering and Information Technology (IAEIT), Pune, too brushes aside misconceptions that aerospace/aeronautical engineers get jobs only in airlines or aircraft-manufacturing units. "The aerospace engineers from our institute have joined IT companies and also the knowledge process outsourcing (KPO) companies. IT companies specialising in this domain need aerospace engineers, so do other aircraft-related outsourcing companies," he shares.

Money matters

This field has enough provisions for not only learning but also earning well. The starting salary for a fresher is typically Rs 300,000-500,000 per annum with impressive year-to-year salary hikes.

Dr Patkar elaborates, "As far as remunerations are concerned, the aerospace engineers would get high-end jobs and hence higher salaries than any other discipline of engineering. We could get the jobs to the fresher to the tune of Rs 300,000 per annum. The rise in this discipline is very high in the private sector."

Prof. Sriram agrees, "Government/public-sector companies (HAL, ISRO, etc) offer standard pay packages in the Rs 300,000-500,000 per annum range at the entry level, while private entities may offer one-and-a-half to twice this. With five-ten years of experience, the salary ranges are around one-and-a-half to double of these figures."

The variation depends on several factors like the nature of the organisation

“The compensation package is dependent on the domain knowledge and nature of experience that the person carries—M.M.T. Nambi, vice president-aviation and business consulting at Ramco Systems



the government/public sector, which includes organisations like ISRO, DRDO, HAL, Air India and National Laboratories.

Then there is the private/IT sector, which includes major engineering companies like Ashok Leyland and Godrej—these firms now have divisions dealing with the aerospace industry and are also recruiting. Earlier, there were small IT companies that specialised in analysis jobs for MNC clients like Boeing and Airbus, but now IT majors like TCS, Wipro and Ramco have also got into this field.

The third category can be MNCs as several multinational firms have set up their operations in India. GE has a large operation in Bengaluru and is probably the largest in this group. Others include Honeywell, Safran, Cessna, Volvo, Airbus, and Boeing-specialised software companies like Ansys & Fluent, Fluidyn and Cd-Adapco.

operated safely.

"Though aircraft maintenance is a separate specialisation and requires qualifying through licensing exams conducted by the Directorate General of Civil Aviation (DGCA), aeronautical engineers would have covered all the theory portions of these exams and can take on these exams after some practical training."

IT companies like TCS, Infosys and Wipro regularly recruit aerospace engineers. In addition, aerospace engineers can find jobs in automobile, construction and manufacturing companies. Aerospace students even find jobs in finance and oil MNCs. Most of these non-core jobs earn much higher pay and are preferred by the students, particularly undergraduate students.

Presently, only a few companies in India work in the field of aeronautics. As a consequence, the number of jobs in the core sector is relatively less. Also,